Do You Know?

Electric motors are used to "actuate" something in your robot: its wheels, legs, tracks, arms, fingers, sensor turrets, or weapon systems. There are literally dozens of types of electric motors (and many more if you count gasoline and other fuelled engines), but for amateur robotics, the choice comes down to these three:

In a stepping motor, applying power causes the shaft to rotate a few degrees, then stop. Continuous rotation of the shaft requires that the power be pulsed to the motor. As with continuous DC motors, there are sub-types of stepping motors. Permanent magnet steppers are the ones you'll likely encounter, and they are also the easiest to use.

Both these stator and rotor fields develop a torque. Then the rotor is rotating in the same direction as the rotating magnetic field. In an induction motor, the rotor speed is always less than the synchronous speed.

8.5.2 Construction of 3 Phase Induction Motor

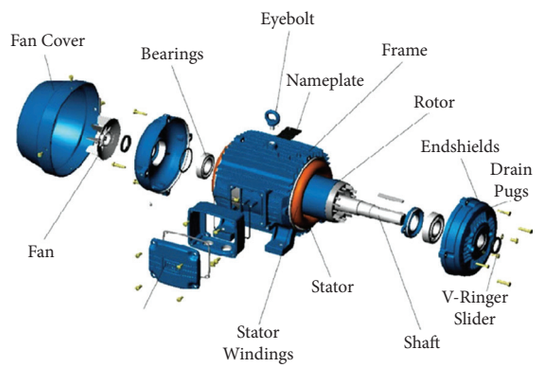


Fig 8.9 *Parts of 3 Phase Induction Motor*

Stator

The stator is made up of number of stamping, with alternate slot and tooth. Stampings are insulated from each other. More number of stampings are stamped together to build the stator core. The stator core is fitted in a casted or fabricated steel frame. The slots in the stator, houses the three phase winding and produces the required rotating magnetic field.

Rotor

Two types of rotor are used in Induction motors. They are:

a. Squirrel cage rotor
b. Slip ring (or) Wound rotor

a) Squirrel Cage Rotor

This is made up of a cylindrical laminated core with slots to carry the rotor conductors. The rotor conductors are heavy bars of copper or aluminum, short circuited in both ends by end rings. The entire rotor resistance is very small. No supply is given or taken. Motors having such type of rotors are extremely rugged in construction. The majority of induction motors are cage rotors.

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